
Your Full Project Title Here

Author One Author Two Author Three Author Four

1. Contributions

Our project compares linear, regularized, and ensemble methods on the Kaggle House Prices dataset. We started from a plain linear regression and ended with a stacked ensemble of Lasso, Gradient Boosting, and XGBoost. We did not chase the lowest RMSE only. A lot of time went into figuring out why each model failed before we moved to the next one. The OLS baseline, for example, gave huge weights to sparse one-hot features. Once we understood that, the move to Lasso and Ridge made sense, and the same kind of reasoning led us to the tree-based models later on.

The full list of models we built and tuned is: OLS as a baseline; Lasso and Ridge with cross-validated regularization; Gradient Boosting with both grid search and randomized search; XGBoost; a weighted hybrid of Lasso and Gradient Boosting; and three stacking setups, with the final one being Lasso + Gradient Boosting + XGBoost. We also compared raw features, `StandardScaler`, and the Yeo–Johnson transform, and ran validation curves on six gradient boosting hyperparameters to see how stable the tuned configuration really was. A lot of the work in the Process section (the `Condition2.PosN` outlier diagnosis, the leaf/split stress testing, etc.) does not show up in the final RMSE table but took significant time, so we want to flag it here.

We used the Kaggle *House Prices: Advanced Regression Techniques* dataset and did not collect anything ourselves. The link to the dataset and our code is in the repository.¹ The repository has a single notebook plus a short list of commands in the README to reproduce all the numbers in the report.

For libraries, scikit-learn did most of the heavy lifting (cross-validation, hyperparameter search, the stacking and voting regressors). XGBoost was used for the boosting model, and pandas, NumPy, SciPy, Matplotlib, and Seaborn for the rest. Two public Kaggle notebooks (jesucristo, 2017; mviola, 2018) helped us with some of the feature-engineering choices.

¹Dataset: <https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques>.
Code: <https://github.com/jwguanusc/CSCI567-FINAL>.

2. AI Usage

We used AI tools (mostly Claude) during the project. How much we relied on them depended on what we were doing.

We did not use AI much for ideation. The general direction – comparing linear, regularized, and ensemble models on a tabular regression task – came from the course content and from looking at top Kaggle solutions. The choice of which models to add at which stage was ours. We sometimes asked AI for a quick second opinion (for example, whether stacking on top of a strong gradient boosting model was worth trying), but we did not let it pick the direction.

Coding was where AI helped the most. A lot of the `scikit-learn` setup is repetitive – `KFold` splits, `GridSearchCV` parameter grids, stacking pipelines, and `matplotlib` plots like the validation curves and feature importance charts. AI was a fast way to get the first version of those, which we then read through, ran, and tweaked to fit our pipeline. We also used it to figure out a few `scikit-learn` error messages and parameter combinations that did not behave as we expected.

For literature review we did not use AI at all. The references in this paper were found through the course materials, the Kaggle forum, and following citations from one paper to another, and we read them ourselves. We were aware of the issue with hallucinated citations, so we manually checked each reference against the actual source.

Sometimes we used AI as a quick reference when a concept was not clear, for example why `StandardScaler` alone is not enough when the features are heavily skewed, or how out-of-fold predictions feed into a stacking ensemble. These were starting points – the actual choices in our pipeline came from our own experiments.

For writing, AI was used to polish the language of paragraphs we had already drafted, and to fix awkward phrasing. The numbers, the figures, and the analysis itself are ours, and we take responsibility for the contents of this report.